

46-976 Financial Optimization, Project
Team formation due on September 24
Main deliverables due on October 2 (preliminary version)
and noon on October 5 (final version)

The goal of this project is to construct portfolios using the universe of stocks in the CRSP database. You will use the universe of stocks in the CRSP database for estimation and for out-of-sample testing. More precisely, you will use the CRSP monthly stock file (`msf_v2`) available via WRDS. If needed, you may also use any of the other data available in WRDS. You will also need market return and risk-free data. The market return and risk-free data can be downloaded from Kenneth French's data library. For your convenience, a CSV file named `MktRf.csv` containing market return and risk-free data (through July 2026) is posted on Canvas. Please note that the scale of the returns in the CRSP data and in Kenneth French's data differ: the latter are returns in *percentage points*.

The two deliverables of this project (preliminary version due on October 2, final version due on October 5) will be based on your work on one of the four possible topics described below. These two deliverables will be a slide deck of your project highlights for a potential 10-minute presentation and a project demo code that any of your peers in the class should be able to use to replicate your results.

You may work in a group of at most four people. The first step in this assignment (worth 2 points) is to form a team by September 24. Please use the link provided on Canvas.

Peer review (5 points, due at 8am on Oct 6)

Each student in the class will be assigned as a peer reviewer for the deliverables (highlights and demo code) of another team. The reviewer will be responsible for assessing the main strengths and weaknesses of each of these two deliverables. The reviewer is expected to provide an assessment and some brief constructive feedback in a professional and timely fashion. Each student in the class will also be asked to cast a vote to help select a team from New York and a team from Pittsburgh to present their highlights to the class during our last class meeting.

It is critical that each peer reviewer adheres to the following timeline:

Read the project highlights and run the demo code of the team under review, and provide a rough assessment (good/fair/poor) and feedback by 8am on Tuesday October 6, 2026.

In addition, each student will be asked to cast a vote for their favorite team from New York and their favorite team from Pittsburgh (excluding their own team) again by 8am on Tuesday October 6, 2026.

Highlights and demo (18 points, due on Oct 2 and Oct 5)

The main project deliverables are based on your work on one of the four topics described in the sections below. To help you avoid submission glitches and to help your peers with their reviews, please submit a preliminary version of your deliverables by midnight on October 2 (worth 3 points). The final version of your deliverables is due on October 5 (worth 15 points). The two specific deliverables are as follows:

- **Project highlights:** A ten-page (or shorter) anonymous slide deck (pdf format) for a potential ten-minute presentation of your project. A few teams will be selected and asked to present their project highlights on the last day of classes.
- **Project demo code:** an anonymous demo code so that your peers, TA, or instructor class can replicate your results. Please assume that your peer reviewers will be able to use their WRDS accounts and will have copies of the `MktRf.csv` file. Ideally your demo should be a single Python notebook. If you need more code and it is cumbersome to include it in your notebook, you can also include one additional Python file. If you need additional data like returns of an index, you may also include one additional CSV file. If you submit multiple files, they should be compressed in a single zip file.

The topics require some steps similar to the ones you have seen in the third exercises in Homework 3 and Homework 4. The different topics will require some extensions and may also provide some opportunities to be creative.

Data and format for the four suggested topics

Each of the topics outlined below fits the following format:

1. You will work with the CRSP monthly stock file (`msf.v2`). You may also need market return and risk-free data.
2. The implementation and backtesting of your models should generally proceed as follows:
 - (a) Select some test period (`startYear`, `endYear`), length T of “in-sample” data history for estimation purposes, and size n of the investment universe. Some reasonable choices are `startYear` = 1995, `endYear` = 2025, T = 60 months, and n = 500 stocks respectively. You may want to report results for different test periods that span different kinds of economic conditions.
 - (b) At the start date estimate the parameters of your models based on the previous T months and compute the relevant portfolios. Then observe and record the performance of these portfolios over the next month. This is your first out-of-sample observation.
 - (c) Repeat the estimation, portfolio construction, and out-of-sample computations for the remaining available dates up to the end date. In principle it would be sensible to estimate parameters and compute portfolios every month. However, you may choose to do it less frequently depending on your time constraints and other considerations of the topic you have chosen. This is a judgement call you will need to make.
 - (d) Summarize and report your out-of-sample results. I recommend summary statistics and cumulative return charts.
3. The four suggested topics require estimation of covariance matrices, betas, and/or stock volatilities. To that end, you may use one or more of the approaches described in the last section of this document instead or in addition to the single-factor covariance estimation described in the third exercise of Homework 3. Feel free to explore other more sophisticated approaches if you wish.

Suggested topics

Your team should choose among the following four suggested topics (the remaining pages of this document provide more details on each):

1. Minimum Risk Portfolios via Various Covariance Estimators
2. Construction of an Index Fund
3. Comparison of Diversification Approaches
4. Active Portfolio Management

The order of the above topics reflects my perception of their difficulty. The steps in the suggested topics should be considered as general guidelines. Feel free to adapt them. That is, you may decide to omit, modify, or add some steps.

If your team wishes to pursue a topic that is completely different from any of the above four topics, and/or use different data, please let me know by submitting your proposal directly to me by 6pm on September 24, 2026. If the proposed project is interesting and comparable in terms of effort and educational value to those suggested in this document, I will most likely agree to let your team work on it. If it is not, I will ask you to choose one of the suggested topics.

First Suggested Topic: Minimum Risk Portfolios via Various Covariance Estimators

Goal: Construct fully-invested, long-only, minimum-risk portfolios via various approaches to covariance estimation. Consider some of the following possible covariance estimates: single factor with and without shrinkage, constant correlation, sample covariance with and without shrinkage (see the last page of this document for estimation approaches). If you wish, you may also explore other estimation approaches, e.g., multiple factor models and/or more sophisticated shrinkage techniques for covariance estimation as those described in the articles posted on Canvas.

In addition to the natural optimization model for fully-invested long only minimum-risk portfolio, you may choose to explore some extensions to manage risk and/or turnover in your portfolio. In particular, to avoid unintentional portfolio concentration on a few stocks, you could include upper bound constraints, e.g., 3% or 5% on each individual holding.

Compare the out-of-sample performance of your minimum-risk portfolios with one or both of the following portfolios:

1. Value (capitalization) weighted portfolio in your investment universe
2. Equally weighted portfolio in your investment universe

Compare also the turnover of your portfolio with the turnover of the above portfolios.

Prepare summary statistics and cumulative return charts of your results with a format similar to those in Exhibit 1 and Exhibit 4 of the following article:

Clarke, de Silva, and Thorley, “Minimum-Variance Portfolios in the U.S. Equity Market,” *Journal of Portfolio Management*, Fall 2006.

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Second Suggested Topic: Construction of an Index Fund

Goal: Construct a small fund to track a large market index. More precisely, your goal is to construct funds with $k \ll n$ stocks that track as closely as possible an index in a universe of n stocks. Two possible choices of indexes are the following benchmark portfolios of the universe comprised of the n largest capitalization stocks available in CRSP each month:

1. Value (capitalization) weighted portfolio
2. Equally weighted portfolio

For the construction of your index fund, you are advised to perform the following steps:

1. Use some kind of clustering or variable selection approach to select the set of k assets that will be included in the tracking portfolio. A possible criterion for clustering is a distance based on the correlation matrix.
2. Assign portfolio weights to the selected assets. A possible approach is to mimic the weighting method used in the index. For example, for a market-value weighted index, assign the weight of each selected stock according to the market value of all the stocks that it represents. You can also attempt to replicate various attributes of the benchmark portfolio, such as beta (relative to the benchmark) equal to one, exposure to particular sectors (subsets of the universe), etc. You can also use an optimization model, e.g., minimization of tracking error, to select the portfolio weights.

In addition to the above steps, you may also explore some extension to your model to manage turnover.

Compare the out-of-sample performance of your index fund portfolios for both the value weighted and equally weighted benchmarks, and for various values of k , for instance $k = \lfloor n/20 \rfloor, \lfloor n/10 \rfloor, \lfloor n/5 \rfloor$.

Third Suggested Topic: Comparison of Diversification Approaches

Goal: Construct fully-invested long-only portfolios via various approaches to diversification: minimum risk, maximum diversification, risk parity. You could also include hierarchical risk parity if you wish. For ease of implementation, I advise you to use a single factor model for covariance estimation as in the third exercise of Homework 3. In particular, that choice of covariance matrix makes the computation of minimum risk, maximum diversification, and risk parity portfolios easier and faster. (Recall Exercise 2 in Homework 4.) However, feel free to use other covariance estimation approaches.

Compare the out-of-sample performance of the above diversification approaches (minimum risk, maximum diversification, risk parity) with the following two ad-hoc approaches:

1. Value (capitalization) weighted portfolio
2. Equally weighted portfolio

Prepare summary statistics of your results with a format similar to that in Exhibit 1 of the following article:

Clarke, de Silva, and Thorley, “Risk Parity, Maximum Diversification, and Minimum Variance: *An Analytic Perspective*,” Journal of Portfolio Management, Spring 2013.

Include also cumulative return charts with a format similar to that in Exhibit 4 of the following article:

Clarke, de Silva, and Thorley, “Minimum-Variance Portfolios in the U.S. Equity Market,” Journal of Portfolio Management, Fall 2006.

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Fourth Suggested Topic: Active Portfolio Management

Goal: Construct a long-only or a long-short investment strategy that outperforms a benchmark in the CRSP investment universe. Please note that this topic is much more open and potentially challenging. This topic relates to some of the material that will be covered in our MSCF Asset Management course next mini.

The following is a possible approach to construct an investment strategy (feel free to be as creative as you wish):

- Explore stock characteristics (such as size, momentum, fundamentals, etc) that are potential indicators of return. The **Compustat** database available in WRDS could come in handy.
- Once you have identified promising characteristics, e.g., “small stocks tend to do better than large stocks”, you will use those characteristics to construct your investment strategy. Typically you proceed as follows:
 - At time t , use the characteristics to rank the stocks using only information available up to time t .
 - Use your ranking to screen your stocks and thus select what stocks to overweight and underweight relative to the benchmark. In other words, convert your ranking to some kind of alpha measure.
 - Choose a rule to construct a portfolio at time t balancing risk and potential for performance.
 - Hold that portfolio until time $t + 1$. Then repeat the above steps.

The ideal active investment strategy (the holy grail of a hedge fund manager) is one that delivers returns with positive premium that are uncorrelated with the benchmark. That is, a strategy with positive alpha and zero beta. This is an extremely difficult (if not impossible) goal. Instead, you are advised to do your best to find an investment strategy that delivers positive alpha with a Sharpe ratio comparable to (or better than) that of the benchmark.

In your highlights, make sure you describe your investment strategy. Include backtests of your strategy over different time horizons. Summarize and report your results the best you can. Think about your highlights are the marketing materials you would present to potential investors.

Estimation of covariance matrix, betas, and volatilities

Suppose the time-series data of stock excess returns $\{\mathbf{r}_t \in \mathbb{R}^n : t = 1, \dots, T\}$ and benchmark excess returns $\{r_{M,t} : t = 1, \dots, T\}$ are available at time $T + 1$ and we want to use this data to estimate $\mathbf{V} = \text{cov}(\mathbf{r}_{T+1})$ and $\boldsymbol{\beta}$ where each β_i is the beta between the i -th stock and the benchmark (“market”) portfolio. In addition to the estimation of the single-factor covariance matrix described in Homework 3, you could consider the next two alternatives.

Estimation of a constant correlation covariance matrix

Consider a second approach to estimating a covariance matrix, namely a constant correlation model. In this case the covariance matrix is of the form

$$\mathbf{V} = \rho \cdot \boldsymbol{\sigma} \boldsymbol{\sigma}^T + (1 - \rho) \text{Diag}(\boldsymbol{\sigma})^2.$$

Here ρ is the (constant) correlation between any two different stock returns and $\boldsymbol{\sigma}$ is the vector of stock volatilities. To estimate ρ and $\boldsymbol{\sigma}$ we can proceed as follows:

1. Estimate ρ as the average of all sample correlations ρ_{ij} .
2. Estimate each σ_i^2 via sample variances and then shrink the log volatilities $1/3$ towards their average log volatilities.

Although estimates $\hat{\rho}_{ij}$ and $\hat{\sigma}_i^2$ of ρ_{ij} and σ_i^2 can be computed naively one at a time, it may be substantially faster to compute them by relying on matrix operations as explained in the second and third exercises of Homework 3.

Shrinkage of the sample covariance

We next describe yet another approach to estimating the covariance by shrinking the sample covariance matrix towards a structured target matrix. This approach is proposed in the technical appendix of the following paper:

Clarke, de Silva, and Thorley, “Minimum-Variance Portfolios in the U.S. Equity Market,” Journal of Portfolio Management, Fall 2006.

Suppose

$$\hat{\mathbf{V}} := \mathbf{R} \mathbf{R}^T$$

where $\mathbf{R} \in \mathbb{R}^{n \times T}$ is the matrix of historical excess returns, that is,

$$\mathbf{R} = \begin{bmatrix} \mathbf{r}_1 & \cdots & \mathbf{r}_T \end{bmatrix}.$$

The matrix $\hat{\mathbf{V}}$ is a scaled version of the sample covariance matrix if we ignore the sample means. The latter practice is common in empirical studies.

Let $\hat{v}_{ij}$ denote the (i, j) entry of $\hat{\mathbf{V}}$. Construct the shrinkage target $\mathbf{C}$ and shrinkage intensity λ as follows. Let $\mathbf{C}$ be the matrix with diagonal entries all equal to the average of the $\hat{v}_{ii}$ and off diagonal entries equal to the average of the $\hat{v}_{ij}$ for $i \neq j$. The shrinkage intensity λ is the following more complicated function of the spread of variance and covariance values around their averages: $\lambda := \max\{0, \min\{1, \bar{\lambda}\}\}$ where

$$\bar{\lambda} := \frac{\sum_{i,j=1}^n \left(\sum_{t=1}^T r_{it}^2 r_{jt}^2 \right) - \frac{1}{T} \cdot \sum_{i,j=1}^n \left(\sum_{t=1}^T r_{it} r_{jt} \right)^2}{\text{trace}((\hat{\mathbf{V}} - \mathbf{C})^2)}.$$

Finally, consider the following shrunk (scaled) estimate of the covariance matrix:

$$\hat{\mathbf{V}} + \lambda(\mathbf{C} - \hat{\mathbf{V}}).$$